TIN LUONG

AI ENGINEER | LLM APPLICATIONS, RAG & APPLIED ML

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PROFESSIONAL SUMMARY

Early-career AI Engineer specializing in LLM applications, retrieval-augmented generation (RAG), and applied machine learning. Combines strong foundations in analytics data modeling and ETL pipelines with hands-on experience developing source-grounded, traceable AI solutions and quantitative retrieval evaluations.

TECHNICAL SKILLS

AI & LLM Systems: RAG Architectures, Vector Search (FAISS), LangChain, Embeddings (OpenAI), Prompt Design, Retrieval Evaluation Fundamentals, Groundedness & Citation Handling.
Machine Learning: Python, NumPy, Pandas, scikit-learn, TensorFlow / Keras, LSTM, K-Means, PCA, Time-Series Forecasting, Feature Engineering, Regression Metrics (MAE, RMSE, R^2).
Engineering & Cloud: SQL (BigQuery, PostgreSQL), FastAPI, TypeScript, REST APIs, AWS (S3, CloudFront, Lambda, API Gateway, DynamoDB), Git/GitHub, Netlify.
Data & Analytics: Data Modeling, ETL Pipelines, Data Validation, A/B Testing, Looker Studio, KPI Architecture.

WORK EXPERIENCE

Data Analyst | Filum.ai *Nov 2025 – Present*
Onsite

- Engineered automated lead-scoring classification pipelines in BigQuery using customer behavioral attributes and interaction data, increasing sales conversion rates by 15%.
- Architected, validated, and maintained centralized data models across touchpoints, tracking 50+ commercial KPIs to ensure high data integrity for operational and predictive analytics.
- Identified churn signals and feature-adoption patterns across user segments, translating complex quantitative data into actionable recommendations for Product and Sales teams.

AI Engineer Intern | CoverGo *May 2025 – Feb 2026*
Remote

- Supported production RAG workflow development for complex medical insurance document retrieval, focusing on context accuracy and source grounding.
- Constructed structured validation datasets comprising 200+ Q&A pairs to benchmark model outputs, identify hallucinations, and track domain-specific exceptions.
- Developed Python and API-based data retrieval automation scripts, reducing manual QA review time by 25% and cutting information retrieval response time.

Data Analyst Intern | MoMo (M_Service) *Nov 2024 – Apr 2025*
Onsite – Ho Chi Minh City

- Architected downstream analytics data schemas for an AI conversational chatbot, structuring user intent logs, session metadata, and dialogue turns for scalable reporting.
- Analyzed dialogue fallbacks, session drop-offs, and A/B test variants across conversational features, contributing to a 10% lift in chatbot user engagement.
- Maintained 2 production data pipelines processing 1M+ rows daily with 99.95% uptime, ensuring clean feature extraction and data integrity.

FEATURED AI & ML PROJECTS

BJJ Research Assistant | Python, LangChain, FAISS, OpenAI API

- Architected a domain-specific RAG assistant parsing martial arts literature with page-aware chunking and OpenAI vector embeddings.
- Implemented FAISS vector search with top-5 cosine similarity retrieval, complete with precise source citations and fallback logic for insufficient context evidence.

Melbourne Weather Lab | Python, Keras, scikit-learn, TypeScript

- Engineered an automated data pipeline processing historical NOAA GHCN-Daily records for multi-task climate analysis.
- Trained a 30-day rainfall LSTM sequence model, a temperature regression model (MAE 3.67°C, RMSE 4.69°C, R^2=0.488), and K-Means weather cluster groupings.
- Exported optimized models for client-side browser inference paired with an interactive TypeScript dashboard.

Photo Album Cloud Architecture | AWS, Serverless, Event-Driven

- Designed a resilient, serverless cloud architecture using AWS S3, CloudFront, API Gateway, Lambda, DynamoDB, and SNS for automated image ingestion and processing.
- Documented system trade-offs between serverless microservices and containerized workflows (ECS/Fargate), incorporating strict IAM access policies.
- **ML Engine: Custom Gradient Descent & Loss Optimization | Python, NumPy, FastAPI, React**
- Implemented linear regression algorithms entirely from scratch in NumPy, deriving gradients and optimization loops for Batch Gradient Descent and SGD.
- Built and benchmarked custom loss functions including MSE, MAE, and robust Huber loss to mitigate the effect of severe price outliers.
- Developed a full-stack inference pipeline serving standardized, one-hot encoded housing features via a FastAPI backend and a React interface.

EDUCATION & CERTIFICATIONS

Bachelor of Data Science | Swinburne University of Technology, Melbourne *2023 – 2026*

- Capstone Project Manager: Led end-to-end analytics delivery, coordinating sprint milestones, stakeholder communications, and final presentations.

Certifications

- Machine Learning Specialization (Andrew Ng / DeepLearning.AI)
- dbt Fundamentals (dbt Labs)